Q1: Let's say we have a 16KiB direct mapped cache which is organized in form of 8-words per block. Also, assume that address field is 64-bits. Find the total size of this cache in bits.

Sol:

8 words per block → $2^m = 8$ → m = 3
For index, 16KiB with 8 words per block → 16KiB/(8*4bytes) → index bits (n) = 512 or 2^9
We can calculate tag field size now,
Tag = 64 – (n + m + 2) = 64 – (9 + 3 + 2) = 50 bits

Single cache line consists of 8*32 + 50 + 1 = 307 bits
We have 512 cache lines → cache size in bit = 512*307 =157184 bits or 19.1875KiB

Q2: For each of the given word addresses, find block address and cache index/block number. Assume we are using the direct mapped cache from Q1.

1. 1200
   Block address = 1200/(8*4) = 37.5 → block address is 37
   Cache index = 37 mod 512 = 37

2. 3308
   Block address = 3308/(8*4) = 103.3 → block address is 103
   Cache index = 103 mod 512 = 103

3. 6784
   Block address = 6784/(8*4) = 212
   Cache index = 212 mod 512 = 212

4. 9604
   Block address = 9604/(8*4) = 300.125 → block address is 300
   Cache index = 300 mod 512 = 300

5. 115200
   Block address = 115200/(8*4) = 3600
   Cache index = 3600 mod 512 = 16